Omar Khodor

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Education

American University of Beirut (AUB), BS in Computer Science — August 2024 – May 2027

- **GPA:** 3.26/4.0
- **Coursework:** Introduction to programming, Data Structures and Algorithms, Discrete Math, AI ethics, Linear algebra, Design and Analysis of Algorithm, Theory of Computation, Software Engineering, and Operating Systems.

Experience

AI Developer Intern | EDM — June 2026 - August 2026

- Built an invoice processing pipeline in Python using **EasyOCR**, **PyTorch**, and **OpenCV** to extract and validate accounting fields with automated review routing.
- Improved gross-total accuracy from 1% to 100% across 414 invoices by identifying the totals block and validating invoice calculations.
- Trained **logistic regression models** with **scikit-learn** on reviewer decisions, achieving 0.99 balanced accuracy, and developed a **SQLite**-based pattern learner for new layouts and languages.

AI Engineering Intern | Catafago — August 2026 - August 2026

- Delivered five internal Python desktop automations (**pandas, Tkinter**) replacing manual spreadsheet work across molecule filtering, supplier reconciliation, forecast enrichment, and PDF table extraction. (GitHub).
- Automated an end-to-end pharmaceutical pricing workflow, replacing per-product manual registry lookups with a single drag-and-drop action producing a finished Excel report. (GitHub).

AI Engineering Intern | Touch Lebanon — July 2026 - July 2026

- Developed a Python AI-powered Excel analysis agent to consolidate network site data and generate an interactive dashboard with site maps, trend charts, and bar graphs.
- Integrated **OpenAI**, **Tkinter**, and **Leaflet**, with secure API key management, to automate data analysis and visualization, reducing approximately an hour of daily manual work.

Security Intern | Touch Lebanon — June 2025 - August 2025

- Conducted phishing simulation campaigns to assess user susceptibility and improve **security awareness**.
- Monitored and analyzed logs using the **ELK Stack** (Elasticsearch, Logstash, Kibana) with Syslog for centralized log collection and visualization.

Projects

AI-Powered Streaming Catalog Analytics Agent

- Developed an **AI analytics agent** that answers natural-language questions about an **Apple TV+ catalog** of 19K titles (genres, release years, IMDb ratings, and regional availability) by generating and running the analysis automatically.
- Automated **report generation** with an **LLM**, producing executive summaries, key trends, data-quality issues, and business recommendations exported to formatted **PDF** reports.

ByteBuy – Hardware Marketplace Platform

- Built a full-stack marketplace platform using **Next.js, Node.js, and React**, implementing RESTful APIs for chat and messaging features (GitHub).
- Designed and integrated a **PostgreSQL** database with secure cookie-based authentication, enabling persistent chat history and real-time-like user interactions.

Bias-Mitigation AutoML Framework

- Extended **Auto-sklearn** by integrating fairness evaluation using Fairlearn, incorporating metrics such as demographic parity and equalized odds to enable threshold-based model selection and bias-aware optimization.
- Developed a filtering mechanism to enforce user-defined fairness constraints during model selection, ensuring compliance while maintaining predictive performance (GitHub).

Extracurricular Activities

AUB Robotics Competition, 2nd Place — March 2025

- Designed and constructed an autonomous robotic boat using an **Arduino** and C programming, integrating control logic and hardware components to improve water circulation and oxygenation in stagnant environments.

Skills and Interests

Programming Tools: Java, Python, C, C++, HTML, CSS, GitHub.

Languages: Arabic (Fluent) English (Fluent), and French (beginner).